



DNA Center Simulator

Individuelle praktische Arbeit



Agenda



Einleitung

Projektumfeld

Mein Auftrag



Realisierung

Verwendete Programme

Gesamtablauf einer Abfrage

Events



Projektmanagement

Projektmethode

Zeitmanagement



Schlusswort

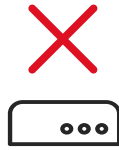
Mein Fazit zum Projekt

Fragen

Demo

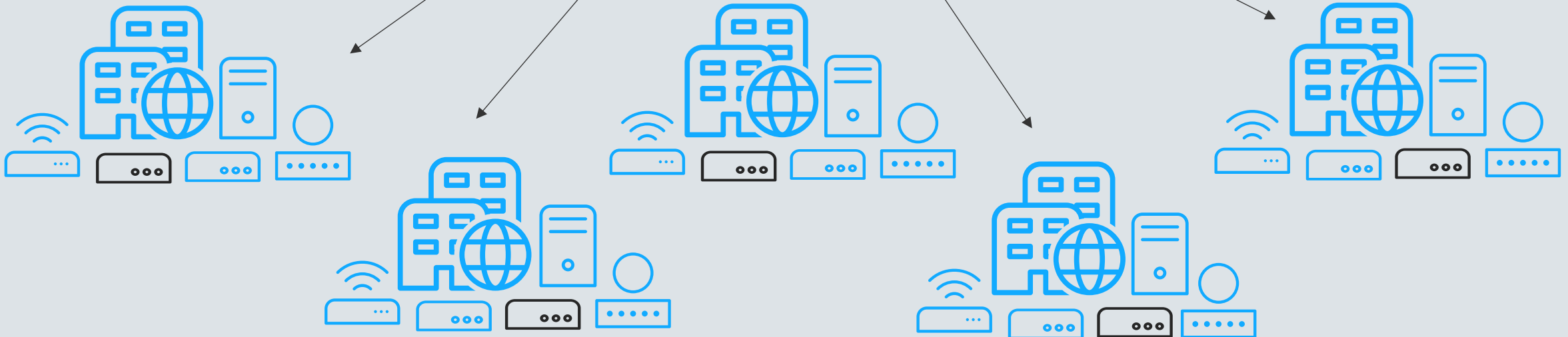


Projektfeld



Supported / Operated Network Service

- Switch
- Router
- Access Points





Mein Auftrag

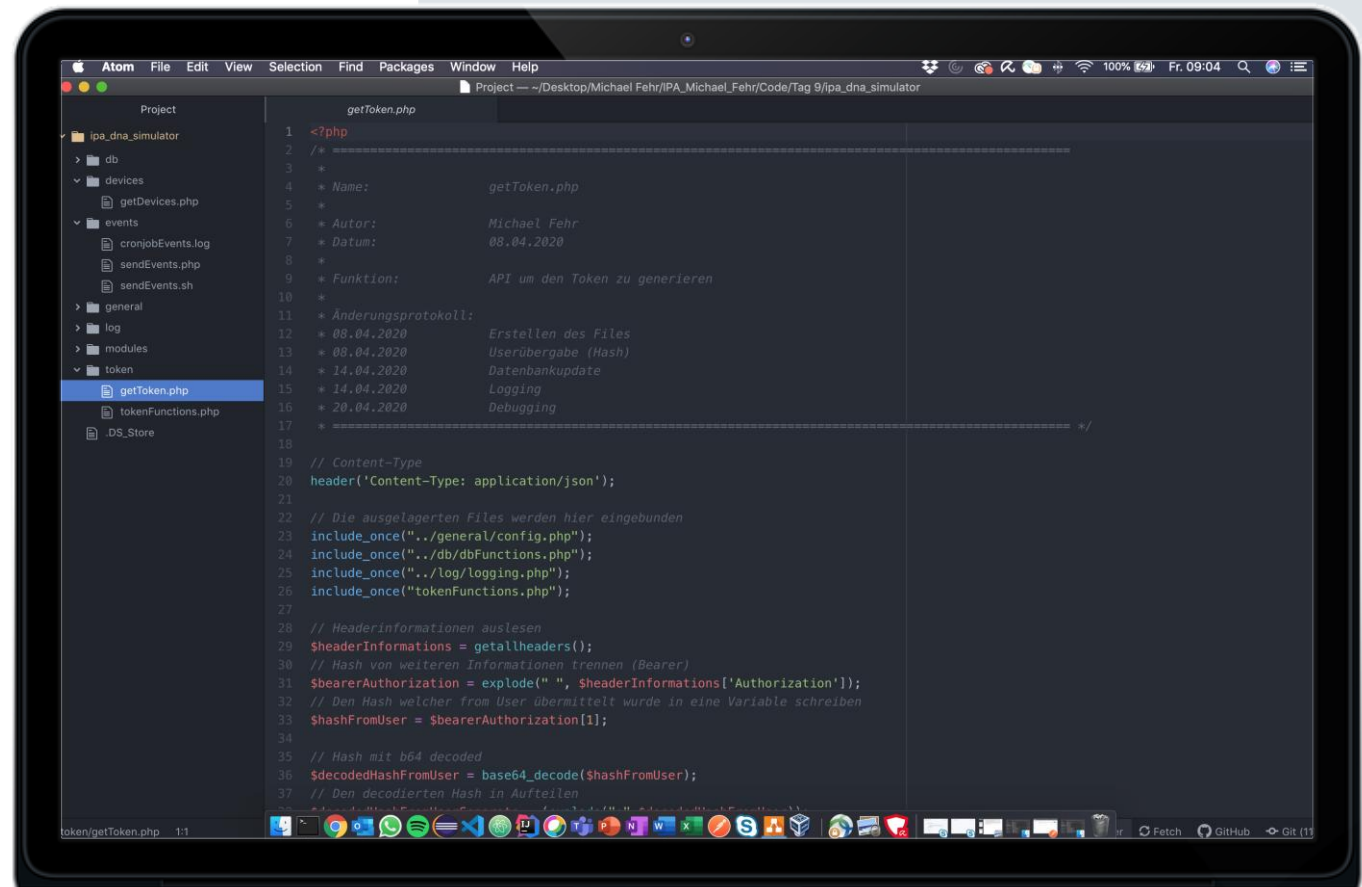
DNA Center Simulator

API Schnittstellen nachbauen

- Authentication API
 - Authentifizierung
- Network Devices
 - Deviceinventar
- Modules
 - Modulinventar
- Events
 - Simulierte Alarmmeldungen

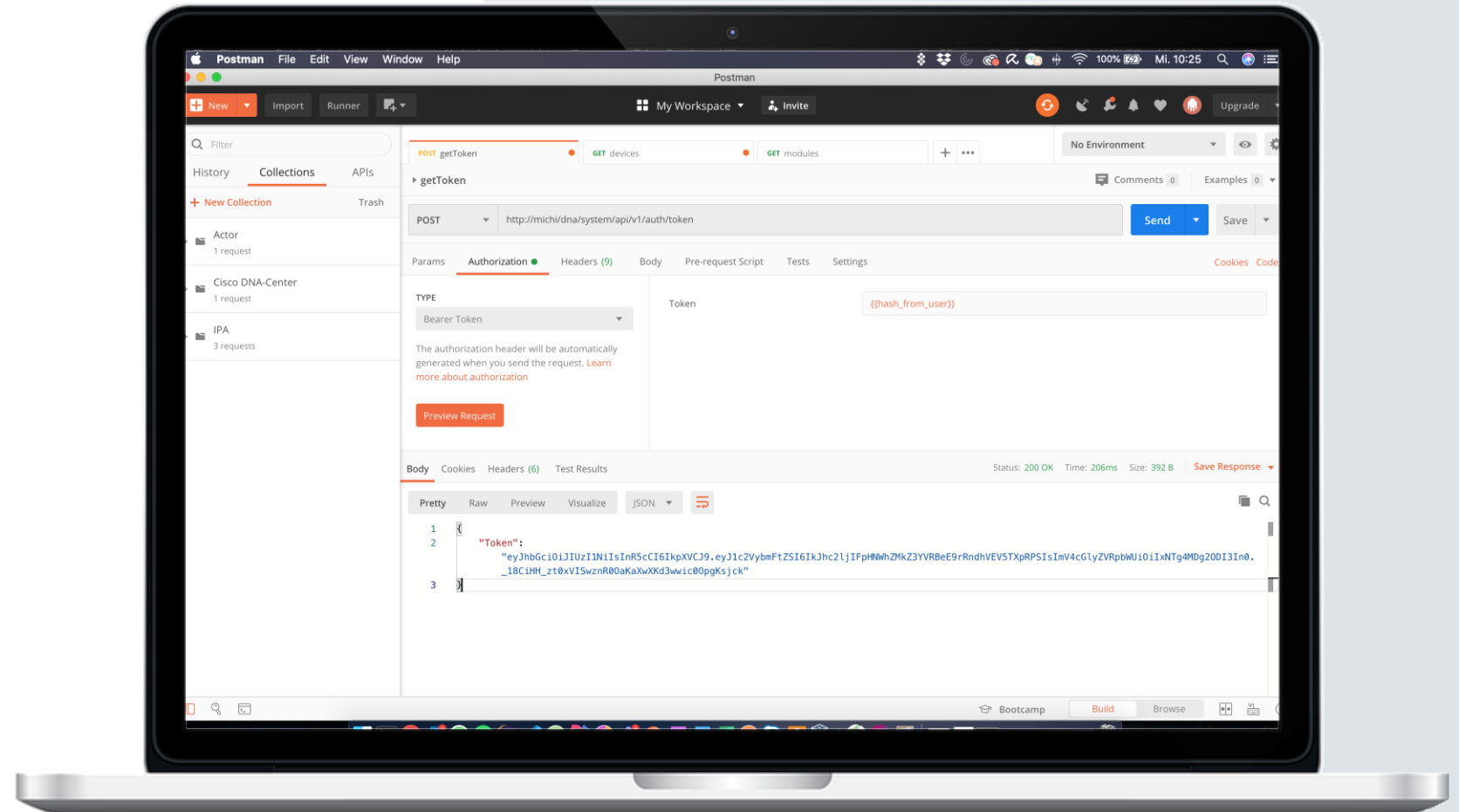
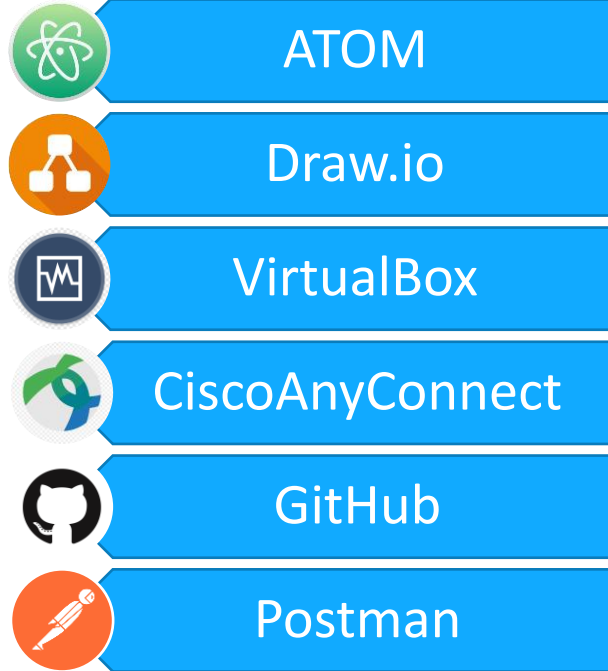
Geräte Virtuell nicht Physisch

Wenige Entscheidungen



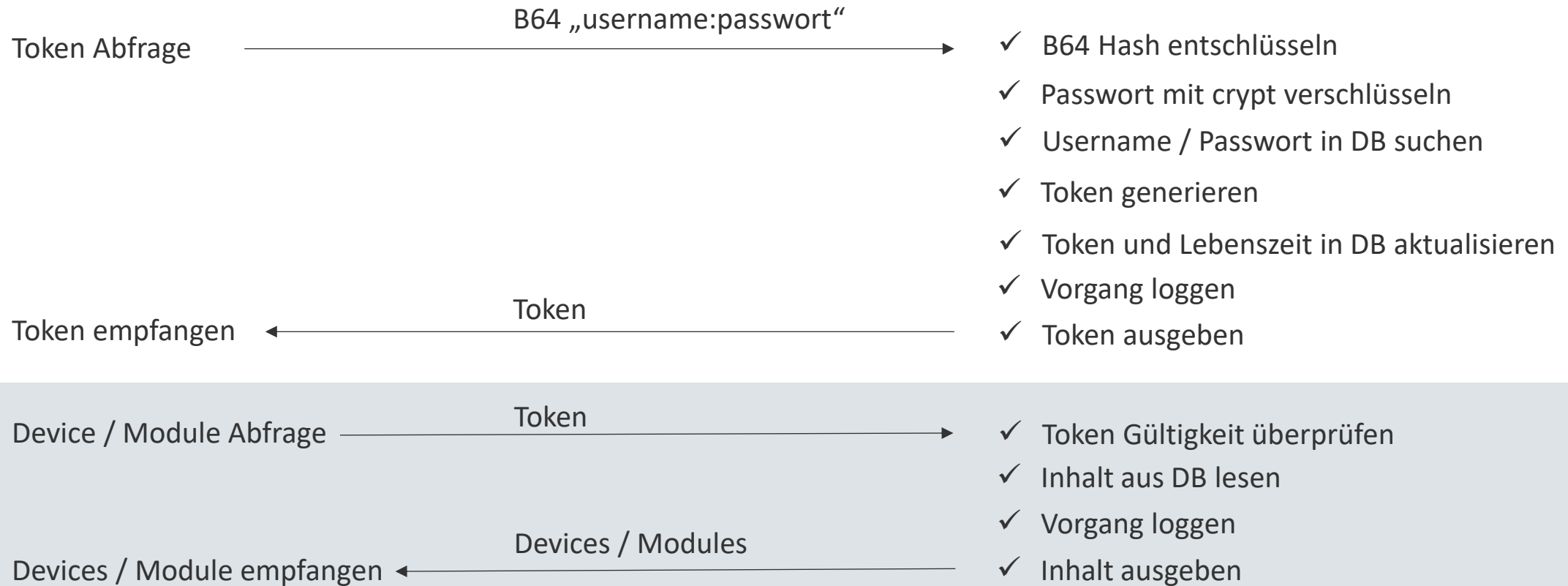


Verwendete Programme





Gesamtablauf





Events

```
mifehr@michi:/var/www/ipa_dna_simulator/events$ php sendEvents.php -c
Dear user,
You have to choose which Eventtype you want to send.
Please choose one of the following Eventtypes by number:

-----
0 | NETWORK-DEVICES-2-152 | WLC Reboot Crash
1 | NETWORK-DEVICES-2-201 | Switch Power Failure
2 | NETWORK-DEVICES-2-202 | Device Reboot Crash
3 | NETWORK-DEVICES-2-204 | Stack Member Running Incompatible Image
4 | NETWORK-DEVICES-2-205 | Switch Fan Failure
5 | NETWORK-DEVICES-3-104 | AP Memory High Utilization
6 | NETWORK-FABRIC_WIRED-1-310 | Fabric Devices Connectivity Control Underlay
7 | NETWORK-NON-FABRIC_WIRED-1-250 | Router unreachable
-----

7
Please write some details about the choosen event:
7
Please provide the severity

-----
1 | Info
2 | Warning
3 | Alert
4 | Critical
-----

1

The event will be sent to http://inventarmichi/receiveEvents
```

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0
Please provide the severity

-----
1 | Info
2 | Warning
3 | Alert
4 | Critical
-----

1

The event will be sent to http://inventarmichi/receiveEvents
```

1. INSERT INTO events (dnaId,dnaEventId,dnaNamespace,dnaName,dnaDescription,dnaType,dnaCategory,dnaDomain,dnaSubDomain,dnaSeverity,dnaTimestamp,dnaDetails,dnaTags,ipaddresses,customerId) VALUES ('instanceId','NETWORK-NON-FABRIC_WIRED-1-250','ASSURANCE','Router unreachable','Device is unreachable','NETWORK','ALERT','domain','subDomain','1','2020-04-20 17:07:58','7','tags','192.168.56.102','0')

||

2. INSERT INTO events (dnaId,dnaEventId,dnaNamespace,dnaName,dnaDescription,dnaType,dnaCategory,dnaDomain,dnaSubDomain,dnaSeverity,dnaTimestamp,dnaDetails,dnaTags,ipaddresses,customerId) VALUES ('instanceId','NETWORK-DEVICES-2-152','ASSURANCE','WLC Reboot Crash','WLC has rebooted due to a hardware or software crash','NETWORK','ERROR','domain','subDomain','1','2020-04-20 17:09:57','0','tags','192.168.56.102','0')

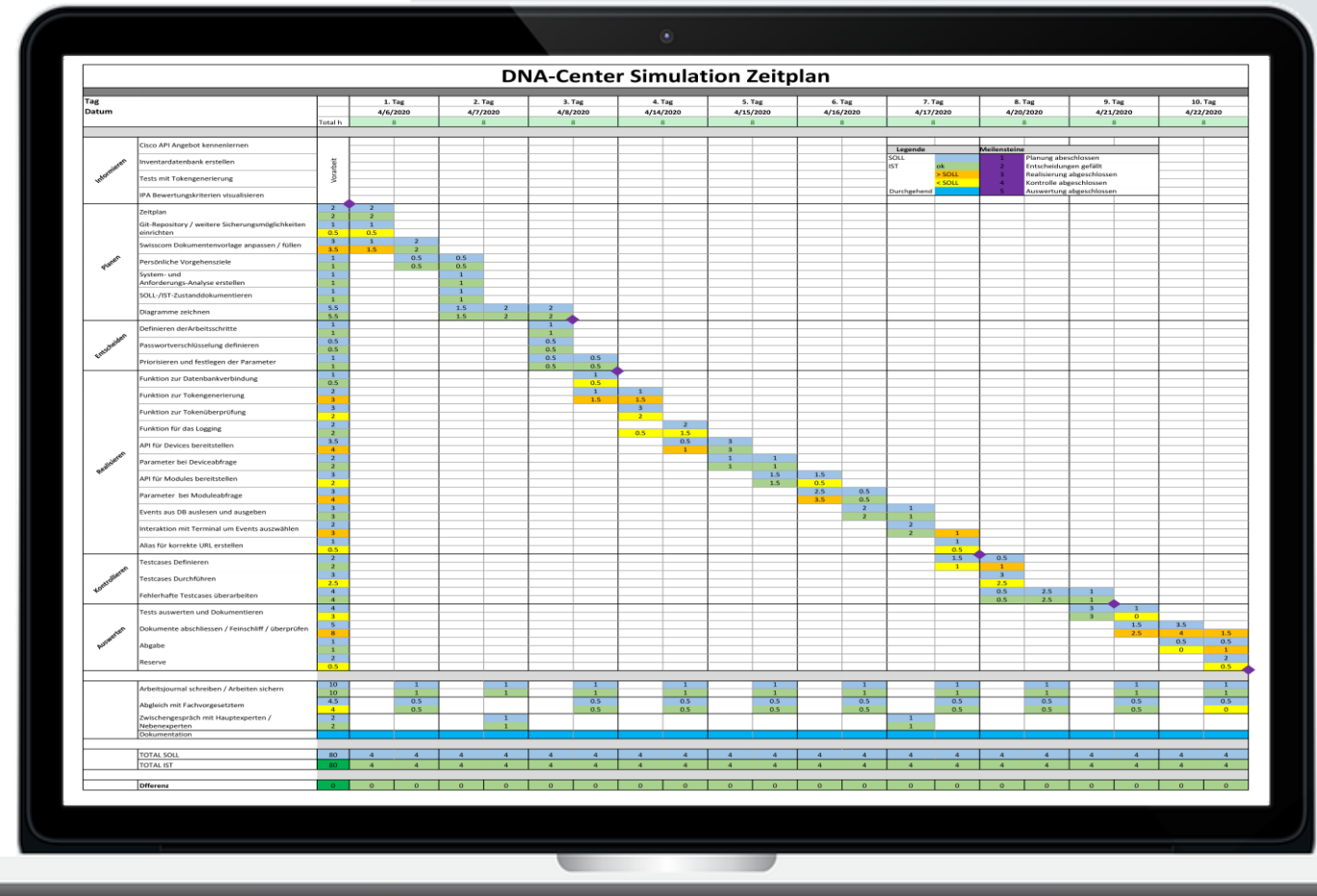


Projektmethode





Zeitmanagement





Schlusswort

Danke!



Erfolge

Funktionstüchtiges Programm
Arbeit unter Stresssituation
Digitale Zusammenarbeit
Zeitplan



Misserfolge

Events unvollständig
Herangehensweise an Probleme



Erkenntnisse

Diagramme im Vorhinein zu zeichnen ist nützlich
Dokumentieren beansprucht viel Zeit
Mehr Reserve einplanen



Demo

01 Token Generierung

Username Passwort enkodieren (dnacapi01:Api1234):

<https://www.base64encode.org/>

Generierter Hash als Variable im Postman eintragen

Abfrage senden

Url = `http://michi/dna/system/api/v1/auth/token`

Erwartetes Resultat:

```
1 {  
2   "Token":  
    "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VybmFtZSI6IkJhc2ljIFpHNWhZMkZ3YVRBeE9rRndhVEV5TXpRPStImV4cGlyZVRpbWUiOiIxNTg4MDg0OTk4In0.ET9pNBW152xE93PM7kPxGWncWmtzC-cDg9eqn4pqM5E"  
3 }
```



Demo

02 Token Generierung mit inkorrekten Userangaben

Inkorrekter Hash als Variable im Postman eintragen

Abfrage senden

Url = `http://michi/dna/system/api/v1/auth/token`

Erwartetes Resultat:

```
1 {  
2   "error": "Authentication has failed. Please provide valid credentials."  
3 }
```



Demo

03 Device Abfrage

Mit korrektem, gültigem Token

Abfrage senden

Url = <http://michi/dna/intent/api/v1/network-device>

Erwartetes Resultat:

```
1 {
2   "response": [
3     {
4       "id": "075e39d1-976a-49e8-ab41-3ee908d504ef",
5       "hostname": "Edge_3.sg.labnet.local",
6       "managementIpAddress": "192.168.8.98",
7       "macAddress": "70:1f:53:01:25:80",
8       "locationName": "None",
9       "serialNumber": "FCW2135L0MU",
10      "location": "None",
11      "collectionStatus": "Managed",
12      "errorCode": "None",
13      "softwareVersion": "16.9.3s",
14      "softwareType": "IOS-XE",
15      "role": "ACCESS",
16      "reachabilityStatus": "Reachable",
17      "associatedWlcIp": null,
18      "type": null,
19      "series": null,
20      "platformId": null,
21
22      "roleSource": "AUTO",
23      "snmpContact": "null",
24      "snmpLocation": "default location",
25      "tagCount": "0",
26      "tunnelUdpPort": "16666",
27      "upTime": "11days 19:28:49.850",
28      "waasDeviceMode": "null"
29    }
30  ],
31  "version": "1.0"
32 }
```



Demo

04 Device Abfrage mit Parameter

Mit korrektem, gültigem Token

Url = <http://michi/dna/intent/api/v1/network-device?id=07a7923e-d3ca-4105-9d9b-48004dbf52be&hostname=ap-west>

Erwartetes Resultat:

```
1 {
2   "response": [
3     {
4       "id": "07a7923e-d3ca-4105-9d9b-48004dbf52be",
5       "hostname": "ap-west",
6       "managementIpAddress": "10.45.254.100",
7       "macAddress": "10:8c:cf:b0:24:70",
8       "locationName": "None",
9       "serialNumber": "FCZ145120L0",
10      "location": "None",
11      "collectionStatus": "Managed",
12      "errorCode": "null",
13      "softwareVersion": "8.5.135.0",
14      "softwareType": "None",
15      "role": "ACCESS",
16      "reachabilityStatus": "Reachable",
17      "associatedWlcIp": "10.252.166.133",
18      "type": "Cisco Catalyst 9300 Switch",
19      "series": "Cisco Catalyst 9300 Series Switches",
20      "platformId": "C9300-24UX",
21      "family": "Switches and Hubs",
22      "description": "24 Port mGig UPOE, Modular Uplinks\r",
23      "apManagerInterfaceIp": "10.44.1.213",
24      "bootDateTime": "null",
25      "collectionInterval": "NA",
26      "errorDescription": "are occurring with this device. Either the SNMP credentials are not correctly provided to controller or the device is responding slow and SNMP timeout is low. If it is a timeout issue, controller will attempt to progressively adjust the timeout in subsequent collection cycles to get device to managed state. User can also run discovery again only for this device using the discovery feature after adjusting the timeout and SNMP credentials as required. Or user can update the timeout and SNMP credentials as required using update credentials",
27      "instanceTenantId": "5b9f5b41056a300088e7bdde",
28      "instanceUuid": "075e39d1-976a-49e8-ab41-3ee908d504ef",
29      "interfaceCount": "0",
30      "inventoryStatusDetail": "NA",
31      "lastUpdateTime": "1583942792872",
32      "lastUpdated": "2020-03-11 16:06:32",
33      "lineCardCount": "0",
34      "lineCardId": null,
35      "memorySize": "NA",
36      "reachabilityFailureReason": "NA",
37      "roleSource": "AUTO",
38      "snmpContact": "null",
39      "snmpLocation": "default location",
40      "tagCount": "0",
41      "tunnelUdpPort": "16666",
42      "upTime": "11days 19:28:49.850",
43      "waasDeviceMode": "null"
44    }
45  ],
46  "version": "1.0"
47 }
```




Demo

05 Modul Abfrage

Mit korrektem, gültigem Token

Url = <http://michi/dna/intent/api/v1/network-device/module?deviceId=b76013b2-71ac-4b6b-afd0-9da7a4f20268>

Erwartetes Resultat:

```
1  {
2    "response": [
3      {
4        "id": "125b4198-ee81-4527-bd4a-1468fb5b456b",
5        "isReportingAlarmsAllowed": "1",
6        "serialNumber": "224-7745-74",
7        "manufacturer": "1235",
8        "typeId": "1",
9        "name": "Switch 1 FRU Uplink Module 1",
10       "description": "8x10G Uplink Module",
11       "vendorEquipmentType": "cevModuleC9300NM8X10G",
12       "partNumber": "C9300-NM-8X",
13       "isFieldReplaceable": "1"
14     },
15     {
16       "id": "e9653aa4-fd41-4bbd-a109-5aee326146f5",
17       "isReportingAlarmsAllowed": "1",
18       "serialNumber": "224-7745-33",
19       "manufacturer": "1236",
20       "typeId": "1",
21       "name": "Switch 1 FRU Uplink Module 1",
22       "description": "8x10G Uplink Module",
23       "vendorEquipmentType": "cevModuleC9300NM8X10G",
24       "partNumber": "C9300-NM-8X",
25       "isFieldReplaceable": "1"
26     }
27   ],
28   "version": "1.0"
29 }
```



Demo

06 Modul Abfrage mit Parameter

Mit korrektem, gültigem Token

Url = <http://michi/dna/intent/api/v1/network-device/module?deviceId=b76013b2-71ac-4b6b-afd0-9da7a4f20268&limit=1>

Erwartetes Resultat:

```
1  {
2    "response": [
3      {
4        "id": "125b4198-ee81-4527-bd4a-1468fb5b456b",
5        "isReportingAlarmsAllowed": "1",
6        "serialNumber": "224-7745-74",
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9        "name": "Switch 1 FRU Uplink Module 1",
10       "description": "8x10G Uplink Module",
11       "vendorEquipmentType": "cevModuleC9300NM8X10G",
12       "partNumber": "C9300-NM-8X",
13       "isFieldReplaceable": "1"
14     }
15   ],
16   "version": "1.0"
17 }
```



Demo

07 Device Abfrage mit falschem Token

Mit ungültigem Token

Url = <http://michi/dna/intent/api/v1/network-device>

Erwartetes Resultat:

```
1 {  
2   "message": "Unauthorized"  
3 }
```



Demo

08 Device Abfrage mit abgelaufenem Token

Mit abgelaufenem Token

Url = <http://michi/dna/intent/api/v1/network-device>

Erwartetes Resultat:

```
1 {  
2   "message": "Unauthorized"  
3 }
```



Demo

09 Events

Automatisch generierter Event

Event Typ 7 manuell erfasst

Erwartetes Resultat:

<i>dnaTimestamp</i> ▼	<i>dnaDetails</i> ▼
2020-04-28 15:24:25	das sind die Details für den Eventtyp 7
2020-04-28 15:23:32	detail 3